

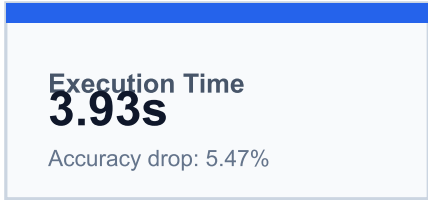
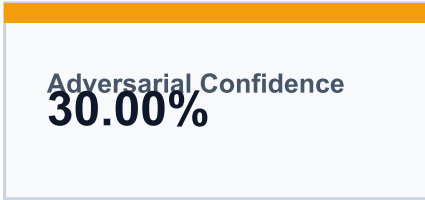
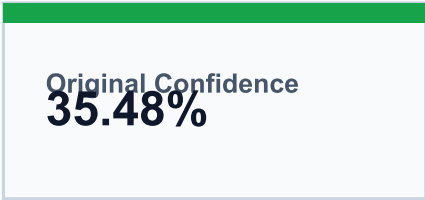
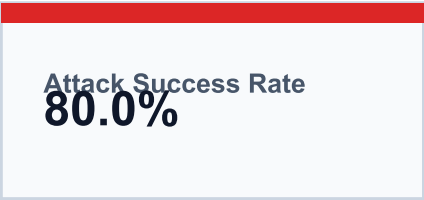
Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: microsoft/resnet-18

Attack Type: FGSM

Images Evaluated: 10



Threat Level Assessment

HIGH RISK

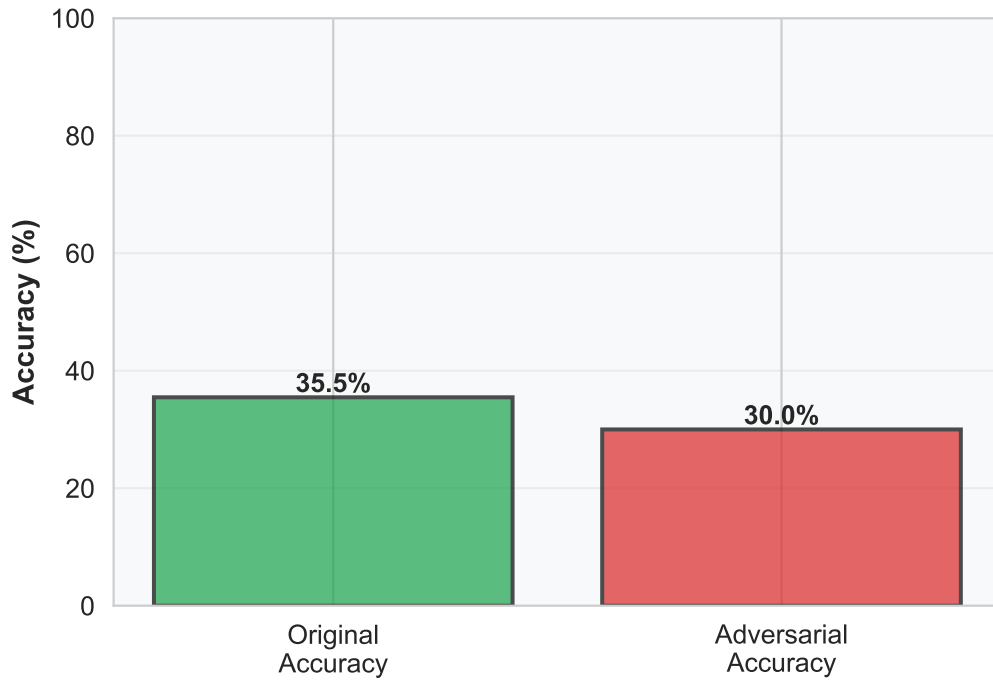
The FGSM attack altered model predictions on 80.0% of the evaluated images. Average confidence shifted from 35.48% to 30.00%, indicating the current robustness posture is high risk.

Assessment Narrative

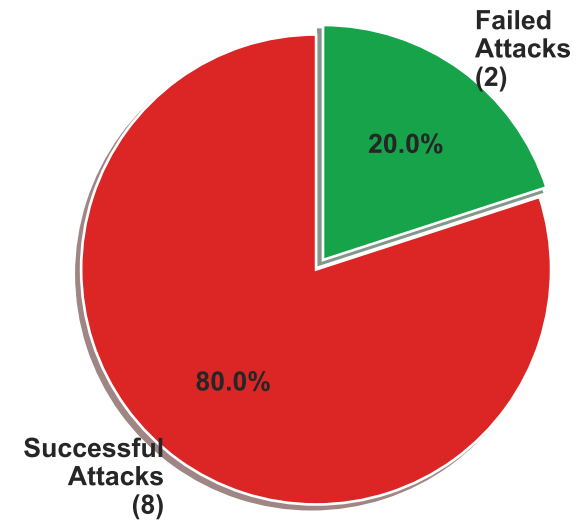
Successfully executed FGSM attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 80.0%. Average original accuracy: 35.48%, Average adversarial accuracy: 30.00%. The attack successfully fooled the model in 8 out of 10 cases.

Detailed Analysis

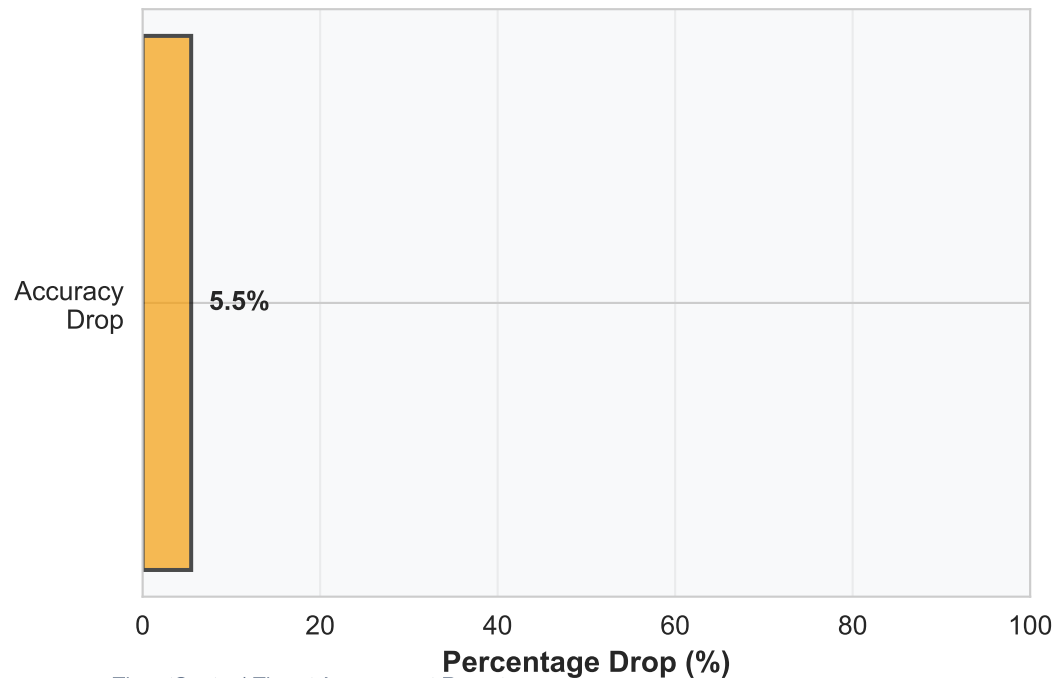
Accuracy Comparison



Attack Success Distribution



Model Robustness Impact



Confidence Comparison (Sample)

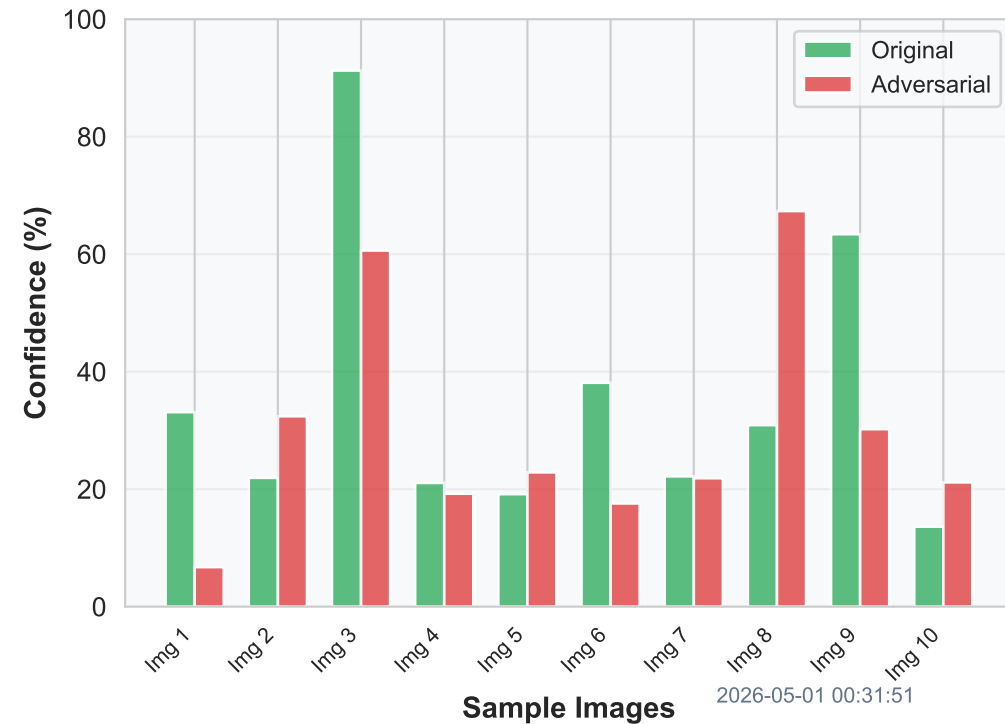


Image Evidence and Predictions

Images 1-4 of 10



Attack Successful

Original image: cello.JPG
Adversarial image: cello.JPG (attacked)
Outcome: Attack Successful
Model predicted as: cardigan
Original confidence: 21.93%
After attack predicted as: shower curtain
Adversarial confidence: 32.41%



Attack Successful

Original image: toy_poodle.JPG
Adversarial image: toy_poodle.JPG (attacked)
Outcome: Attack Successful
Model predicted as: stole
Original confidence: 13.63%
After attack predicted as: spider web, spider's web
Adversarial confidence: 21.16%



Attack Blocked

Original image: electric_guitar.JPG
Adversarial image: electric_guitar.JPG (attacked)
Outcome: Attack Blocked
Model predicted as: electric guitar
Original confidence: 91.26%
After attack predicted as: electric guitar
Adversarial confidence: 60.63%



Attack Successful

Original image: drum.JPG
Adversarial image: drum.JPG (attacked)
Outcome: Attack Successful
Model predicted as: lampshade, lamp shade
Original confidence: 38.12%
After attack predicted as: shower curtain
Adversarial confidence: 17.58%

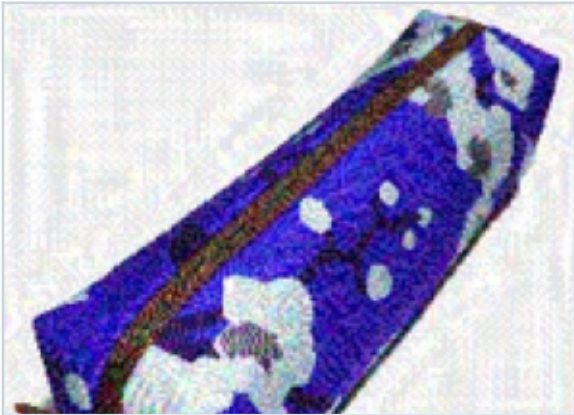
Image Evidence and Predictions

Images 5-8 of 10



Attack Successful

Original image: dishrag.JPEG
Adversarial image: dishrag.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: dishrag, dishcloth
Original confidence: 63.40%
After attack predicted as: prayer rug, prayer mat
Adversarial confidence: 30.22%



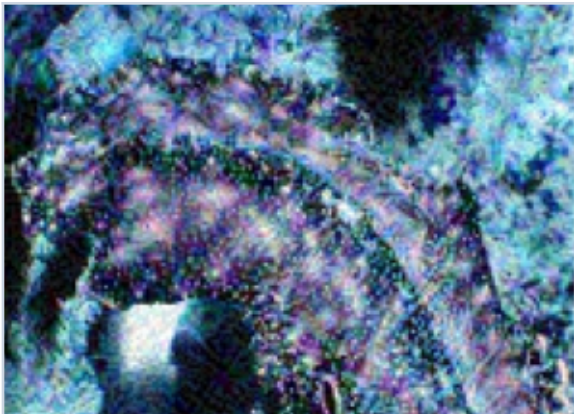
Attack Successful

Original image: pencil_box.JPEG
Adversarial image: pencil_box.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Windsor tie
Original confidence: 30.90%
After attack predicted as: pencil box, pencil case
Adversarial confidence: 67.34%



Attack Successful

Original image: wallet.JPEG
Adversarial image: wallet.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: jigsaw puzzle
Original confidence: 22.19%
After attack predicted as: envelope
Adversarial confidence: 21.87%



Attack Blocked

Original image: flatworm.JPEG
Adversarial image: flatworm.JPEG (attacked)
Outcome: Attack Blocked
Model predicted as: coral reef
Original confidence: 21.07%
After attack predicted as: coral reef
Adversarial confidence: 19.23%

Image Evidence and Predictions

Images 9-10 of 10



Attack Successful

Original image: Norwich_terrier.JPEG

Adversarial image:

Norwich_terrier.JPEG (attacked)

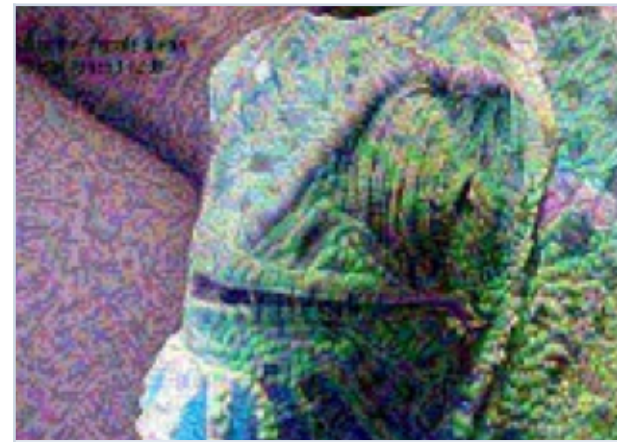
Outcome: Attack Successful

Model predicted as: stole

Original confidence: 19.16%

After attack predicted as: African
chameleon, Chamaeleo chamaeleon

Adversarial confidence: 22.87%



Attack Successful

Original image: African_chameleon.JPEG

Adversarial image:

African_chameleon.JPEG (attacked)

Outcome: Attack Successful

Model predicted as: sock

Original confidence: 33.12%

After attack predicted as: jigsaw
puzzle

Adversarial confidence: 6.74%

Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 80.0%. Average original accuracy: 35.48%, Average adversarial accuracy: 30.00%. The attack successfully fooled the model in 8 out of 10 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques